

Do CNNs Trained on Brain MRI Tumor Classification Rely on Clinically Meaningful Features or Dataset-Specific Artifacts?

CSE 493G1: Deep Learning Final Report

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Abstract

Deep learning models, specifically Convolutional Neural Networks (CNNs) have achieved high accuracy on classifications of brain MRI tumors, but high accuracy does not guarantee that these models are learning features that are clinically meaningful. In this project, we investigate whether a ResNet-18 classifier trained on a brain MRI tumor dataset relies on medically relevant brain structures or relies on dataset-specific spatial artifacts. Using Grad-CAM visualizations we find the baseline model consistently attends to skull boundary regions when it misclassifies glioma tumors as meningioma, aligning with location based shortcut learning. Then, we applied a preprocessing intervention that combined center-cropping and data augmentation to discourage reliance on this spatial shortcut. This intervention model shows that the attention partially shifted toward internal brain tissue in misclassified cases, but overall accuracy decreases from 96.1% to 93.5% which suggests that some of the baseline performance was driven by spurious correlations. These results demonstrate that CNN-based brain MRI classifiers can exploit non-biological artifacts, and that simple preprocessing interventions can partially but not fully mitigate this behavior.

1. Introduction & Motivation

Convolutional Neural Networks (CNNs), have become popular in medical imaging due to their ability to detect complex visual patterns. In magnetic resonance imaging (MRI)'s of the brain, CNNs are used for tasks such as tumor detection and classification. These systems often achieve a high accuracy, which has led to a growing interest in their use for supporting clinical decisions.

However, the high accuracy doesn't guarantee that the model is learning medically meaningful information. Recent research has shown that medical imaging models frequently rely on shortcut learning, where the CNN bases its predictions on correlations driven by misleading, but easy to detect patterns instead of actual clinically meaningful features. Examples of these shortcuts include noise patterns such as image borders, text markers, acquisition artifacts, or differences in image preprocessing between classes.

This presents a major challenge for clinical reliability. A model that appears highly accurate during testing may fail when deployed in the real world if the spurious features it learned are absent or different. Therefore, understanding why a model makes a prediction is just as important as the prediction itself in healthcare applications.

To investigate this issue, our project studies whether CNN brain MRI tumor classifiers focus on clinically relevant brain regions or instead exploit non-biological artifacts. We use explainability techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the image regions influencing model decisions and evaluate whether these regions correspond to anatomically meaningful structures. In

addition, we explore whether simple preprocessing and training interventions can reduce shortcut learning and encourage the model to look at features more relevant to tumors.

2. Related Work

Source One: The Risk of Shortcutting in Deep Learning Algorithms for Medical Imaging Research

This paper discusses how deep learning models in medical imaging can achieve high performance while still relying on unintended or spurious patterns in the dataset instead of medically meaningful features. The paper explains that models learn shortcuts because these correlations are easier for the model to detect compared to medically meaningful traits. The paper also emphasizes that shortcut learning can reduce reliability when models are tested on new data from different hospitals or scanners. The authors continue on to explain that shortcut learning is a major challenge for trustworthy medical AI where robustness and interpretability matter. The overall question it is trying to answer is “Why is shortcut learning dangerous?” This paper is similar to our project because both focus on shortcut learning in medical imaging and investigating whether CNNs rely on clinically meaningful features. This paper is different from our project because it is broader and looks at medical imaging in general, whereas we focus on brain MRI tumor classification specifically. We plan to test preprocessing interventions to see if shortcut learning can be reduced, which is something this paper doesn’t address.

Source Two: Detecting Shortcut Learning for Fair Medical AI Using Shortcut Testing

This paper discusses how medical AI systems can unintentionally exploit spurious correlations and biases in datasets when making predictions. The researchers introduce “shortcut testing” methods that help determine whether a model is relying on shortcuts instead of medically meaningful information. The paper focuses heavily on understanding model behavior and improving fairness and reliability in medical AI systems. The authors emphasize that models can appear accurate while still making decisions for the wrong reasons. The overall question this paper is trying to answer is “How can we detect shortcut learning?” This paper is similar to our project because both investigate shortcut learning and aim to understand why medical imaging models make certain predictions. This paper is different from our project because it focuses more on fairness and testing methodologies for identifying shortcuts, while our project focuses more on explainability using Grad-CAM visualizations. Our project also specifically studies brain MRI tumor classification and tests preprocessing interventions to reduce shortcut learning, while the paper is more generalized.

Source Three: Shortcut Learning in Medical AI Hinders Generalization: Method for Estimating AI Model Generalization Without External Data

This paper explains how medical AI models often fail to generalize because they learn shortcuts that only exist within the training dataset. The authors describe how models may rely on scanner artifacts, background patterns, or hospital-specific imaging characteristics instead of actual disease-related information. The paper argues that this creates problems when models are applied in real clinical settings with different scanners or patient populations. The paper’s main question is: “How does shortcut learning hurt generalization?” This paper is similar to our project because both investigate whether CNNs rely on non-biological features rather than medically meaningful brain structures. This paper is different from our project because it focuses more broadly on generalization issues across datasets and institutions, while our project specifically studies brain MRI tumor classification. Our project also uses Grad-CAM to visually analyze model attention patterns and compare baseline and intervention models.

3. Methods

3.1. Dataset

We used the Sartaj Brain Tumor Classification (MRI) dataset, which sorts brain MRI scans into four categories: (1) glioma, (2) meningioma, (3) pituitary tumor, and (4) no tumor. The images are grayscale axial, coronal, and sagittal slices that vary in resolution and contrast, since they come from different scans rather than a single standardized source.

The dataset splits into separate 2 separate Training and Testing, but the original test folder is small and has known labeling problems, so we merged the two into one pool and re split the data ourselves. From this merged set we drew a 70/15/15 train/validation/test split using a fixed random seed of 42, which kept the partition identical across all three of us and made our results reproducible.

3.2. Preprocessing

Every image is passed through a 4 step pipeline before reaching the model. (1) since the MRIs are grayscale but ResNet-18 needs a three input channel, we loaded each image as a RGB by replicating the single grayscale channel across all three. (2) we resized every image to 224×224 using bilinear interpolation to match the input resolution ResNet-18 was pretrained on. (3) we converted the image to a tensor, which also rescales pixel values from the 0 to 255 integer range down to floats between 0 and 1. (4) we normalized each channel using the standard ImageNet mean and standard deviation so that the pretrained weights would see inputs in the distribution they were trained on.

The intervention model used the same base pipeline plus a few extra transformations to try and reduce shortcut learning. For the intervention model we added a center crop to 180 pixels followed by a resize back to 224×224, which trims away the outer edge of the image where the skull boundary sits. On top of that we applied random horizontal flips, random rotations of up to 15 degrees, and color jitter on brightness and contrast. Section 3.6 explains the reasoning behind these choices.

3.3. Model Architecture

We used ResNet-18 (pretrained on ImageNet), and swapped out its final fully connected layer for one with four outputs to match our four tumor classes. We picked ResNet-18 because its residual connections makes it stable to train and let gradients flow cleanly through the network, which matters since we have a relatively small medical dataset. Because we started from ImageNet weights, it gave us the benefit of transfer learning, so our model already knew useful low-level visual features and only needed to adapt to MRI data. The architecture is also lightweight enough to train quickly on a single GPU, and its clean convolutional structure makes it straightforward to attach Grad-CAM to the final convolutional block.

3.4. Training Setup

We trained both models with cross-entropy loss and the Adam optimizer at a learning rate of 1e-4, for 10 epochs with a batch size of 32. Training ran on a T4 GPU through Google Colab. We also tracked training and validation accuracy at the end of each epoch so we could watch for overfitting and confirm the models were converging, and the curves will be shown in Section 4.3.

3.5. Grad-CAM Visualization

We used Gradient-weighted Class Activation Mapping (Grad-CAM) to see where the model was actually looking at. We wanted to take the score the model assigned to its predicted class and compute how that

score changes with respect to the feature maps in the last convolutional layer (in our case layer4[1].conv2). Then averaging these gradients gives a weight for each feature map that reflects how important it was to the prediction. We then combine the feature maps using those weights and keep only the positive contributions, which produces a coarse spatial heatmap highlighting the regions that pushed the model toward its decision. We then overlaid on the original MRI, and this heatmap shows whether the model focused on tumor tissue inside the brain or on something else, like the skull edge for instance.

3.6. Intervention

Our intervention was designed to take away the spatial shortcut the baseline model relied on. Our Grad-CAM analysis showed that the baseline was using a tumor’s position relative to the skull boundary as a cue for its class, rather than the tumor’s actual appearance. Therefore, we introduced the center crop which directly targets this by cutting off the outer ring of the image, removing much of the skull boundary information the model could lean on. Additionally the augmentations such as random flips and rotations stop the model from tying a class to a fixed location in the frame, and color jitter discourages it from latching onto scanner specific brightness or contrast patterns. Together these transforms force the model to find features inside the brain itself, so that if the model still classifies well, it is doing so for more medically meaningful reasons.

4. Experiments & Results

4.1. Baseline Model Performance

Table 1: Baseline ResNet-18 test set performance (macro-averaged).

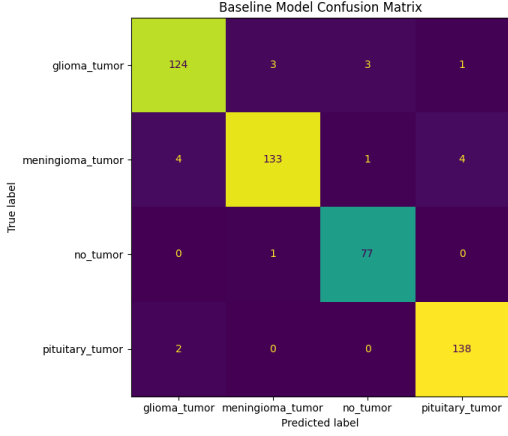
Metric	Baseline	Intervention
Accuracy	0.961	0.935
Precision	0.960	0.938
Recall	0.964	0.936

4.2. Confusion Matrices

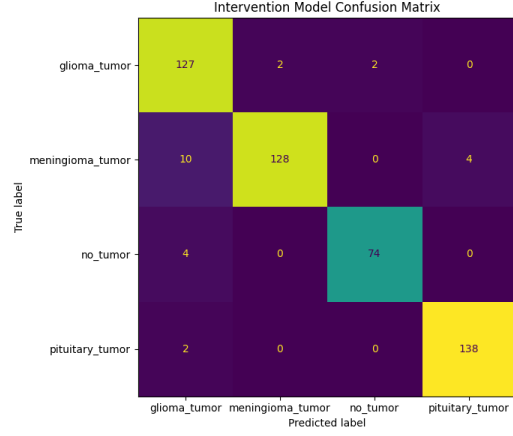
The confusion matrices for both models reveal that the most prevalent error pattern in both cases is the confusion between glioma and meningioma. In the baseline model, 3 glioma tumors were misclassified as meningioma and 4 meningioma tumors were misclassified as glioma.

In the intervention model, this pattern shifts: glioma-to-meningioma confusions decreased slightly ($3 \rightarrow 2$), but meningioma-to-glioma confusions increased substantially ($4 \rightarrow 10$).

All other class pairs are well-classified in both models, with near-perfect accuracy on no_tumor and pituitary_tumor. This confirms that the primary source of error in both models is the glioma–meningioma boundary, and this is consistent with our Grad-CAM finding that the model uses tumor location relative to the skull as a shortcut feature.



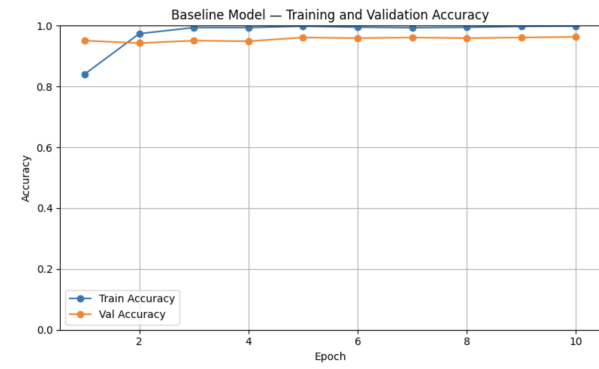
(a) Baseline model.



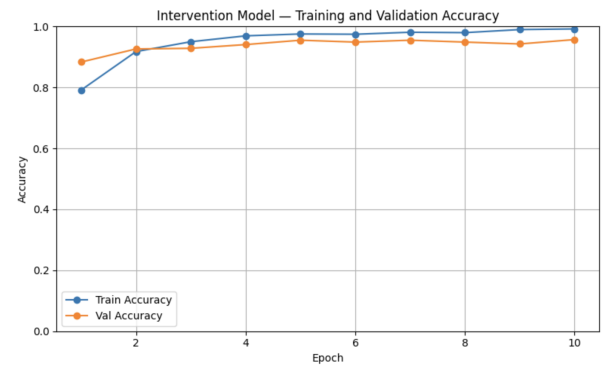
(b) Intervention model.

Figure 1: Confusion matrices for the baseline and intervention ResNet-18 models. The primary error in both models is glioma–meningioma confusion, consistent with location-based shortcut learning.

4.3. Training Curves



(a) Baseline model.



(b) Intervention model.

Figure 2: Training and validation accuracy over 10 epochs for the baseline ResNet-18 (left) and the intervention model (right).

4.4. Grad-CAM Analysis: Baseline Model

A trend that emerges from the Grad-CAM analysis is that, compared to inaccurate classifications, accurate classifications exhibit a more focused attention pattern localized around brain tissue, whereas inaccurate classifications show increased attention to the edges of the brain and the skull. This can be seen from Figure 3 where the correctly classified cases (rows 1-2) show attention that is localized within brain tissue while the incorrectly classified cases (rows 3-4) display an increased amount of attention focused on edges of brain and skull.

This pattern is consistent with location-based shortcut learning. Meningiomas form at the meninges, which are the layers covering the brain, hence appearing near the edges of the skull in MRI images. Instead of learning the distinguishing features of a tumor, the model learns to classify them on the basis of their locations

relative to the skull edge, leading to the misclassification of glioma tumors that appear near the edges of the skull.

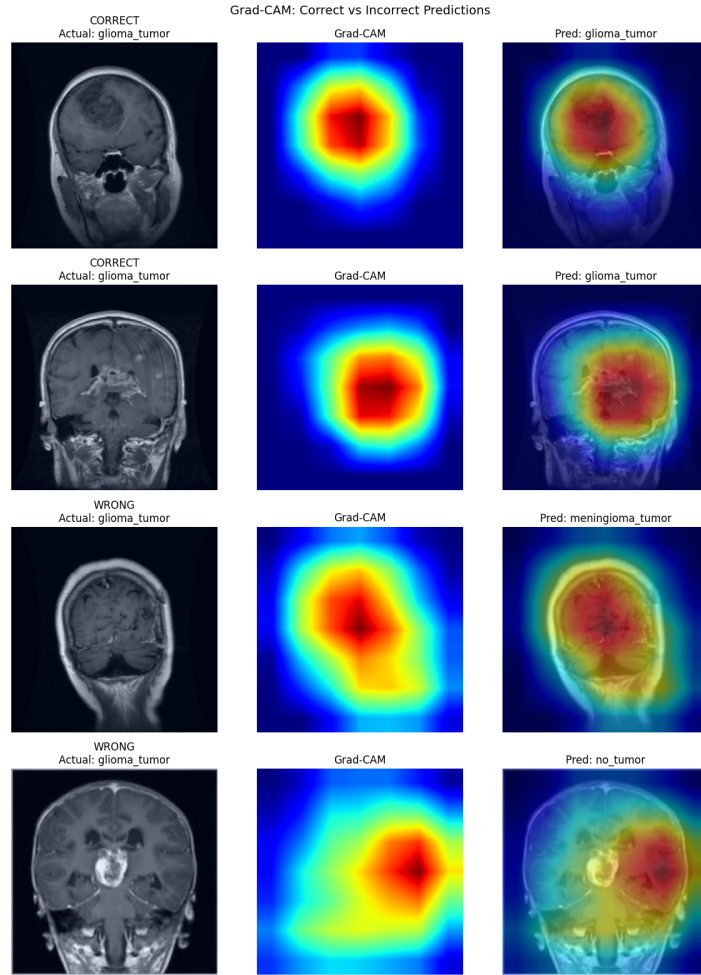


Figure 3: Grad-CAM visualizations for the baseline ResNet-18. Rows 1–2 show correctly classified glioma tumors with attention inside the brain region. Rows 3–4 show glioma tumors misclassified as meningioma, with attention concentrated at the skull boundary — consistent with location-based shortcut learning.

4.5. Grad-CAM Analysis: Intervention Model

The Grad-CAM visualizations for the intervention model show a partially shifted attention pattern compared to the baseline model. Rows 1 and 2 correspond to correctly predicted results. In both cases, attention is concentrated on the inner regions of the brain, similar to the baseline model’s behavior. In the incorrectly predicted results (rows 3 and 4), however, attention is focused on different features. For the meningioma tumor that was incorrectly classified as a glioma (row 3), attention is concentrated on the upper portion of the brain. This suggests that the error stemmed from feature ambiguity rather than location-based shortcut learning, since the skull margin was not used. Similarly, in the fourth row, attention is focused on the lower-central region of the brain, away from the skull edge that appeared prominently in the baseline model’s incorrect predictions.

Although in the baseline model misclassifications consistently exhibited strong attention near the edges

of the skull, the Grad-CAM maps for the intervention suggest a preference for attending to internal brain regions. This indicates that the combination of tight cropping and data augmentation successfully disrupted the model’s reliance on this specific shortcut. The decrease in accuracy from 96.1% to 93.5% further supports this interpretation: once the model lost access to the shortcut, its performance declined, suggesting that a portion of the baseline model’s accuracy was driven by spurious spatial correlations rather than genuine tumor morphology.

This pattern holds systematically across both models as correct predictions consistently show focused internal attention while incorrect predictions show boundary-focused activation.

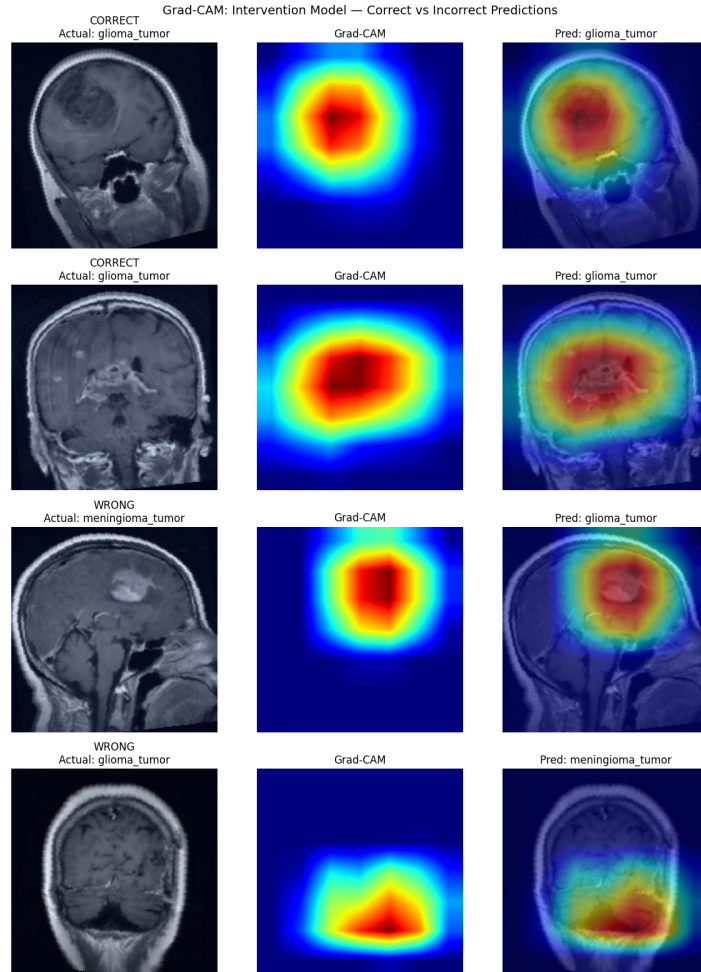


Figure 4: Grad-CAM visualizations for the intervention ResNet-18. Correct predictions (rows 1–2) show attention concentrated on internal brain tissue. Incorrect predictions (rows 3–4) display less consistent skull-boundary focus compared to the baseline, suggesting the intervention partially reduced location-based shortcut learning.

5. Discussion & Conclusion

Our findings suggest the baseline ResNet-18 did rely on shortcut learning since it used a tumor’s position relative to the skull boundary as a stand-in for its class, rather than learning what the tumor actually looked like. This was most obvious in the systematic misclassification of glioma tumors as meningioma, and

Grad-CAM heatmaps consistently showed activation along the skull edge rather than internal tumor tissue. The preprocessing intervention partially disrupted this shortcut, with misclassified images in the intervention model showing attention shifted toward interior brain regions, though at the cost of a small accuracy decrease from 96.1% down to 93.5%. These results highlight a critical gap between classification accuracy and clinical reliability. A model can achieve high test accuracy while making predictions for the wrong reasons, which poses serious risks in clinical deployment.

Several limitations temper these conclusions. Grad-CAM is a post-hoc approximation and does not provide ground truth for what the model has learned; and we don't have a radiologist who verified labels for clinical meaningfulness. The dataset is relatively small, only one architecture was tested, and the intervention transformations we applied may not fully eliminate all forms of shortcut learning present in the data.

This project reinforced that high accuracy alone is not sufficient evidence of a trustworthy medical AI system. Working through the full pipeline, from training and evaluation to interpretability analysis and intervention, gave us a concrete understanding of how shortcut learning manifests in practice and how difficult it is to eliminate fully. We also learned that research findings are rarely clean and our intervention produced mixed Grad-CAM results that required an honest interpretation from our part rather than a straightforward conclusion. Ultimately, this work demonstrates that CNNs trained on brain MRI tumor classification can exploit dataset-specific spatial artifacts, and that simple preprocessing interventions can partially mitigate but not fully resolve this behavior.

References

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